
Software Requirements Specification

for

KR Buddy

Version 1.0 Approved

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Revision History

Name	Date	Reason for Changes	Version
Shuvo	2026-04-02	Initial draft of SRS document	0.1
Sudipta	2026-04-04	Added use case and sequence diagrams	0.5
Shuvo	2026-04-07	Final version submitted	1.0

1. Introduction

1.1 Purpose

This document provides a detailed description of the software requirements for KR Buddy, a web-based platform designed to assist international students in South Korea.

1.2 Document Conventions

This document follows standard IEEE SRS formatting. Headings are numbered hierarchically. Requirements are labeled as REQ-x. All technical terms are defined in Appendix A.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, instructors, and stakeholders of the KR Buddy project. Developers should focus on Sections 3–5. Instructors and evaluators may review all sections for system understanding.

1.4 Scope

KR Buddy is a full-stack web application that provides structured guidance on:

- ARC (Alien Registration Card)
- Visa
- Housing
- Jobs
- Emergency information
- Korean language basics

The system includes a minimal backend for storing user queries.

1.5 Overview

This document includes system features, functional requirements, and diagrams.

2. Overall Description

2.1 Product Perspective

- KR Buddy is a standalone web application designed to assist international students in South Korea.

- The system provides structured guidance on essential topics such as visa, housing, ARC, and emergency services. It integrates a frontend interface with a backend system and database to store user queries.
- The system is not dependent on external enterprise systems but utilizes Supabase as a backend service provider.

2.2 Product Functions

The system allows users to:

- Access guides
- View emergency information
- Learn Korean phrases
- Submit help requests

2.3 User Characteristics

- International students
- Basic computer knowledge
- Non-Korean speakers

2.4 Constraints

- Internet required
- No authentication system
- Minimal backend

2.5 User Documentation

The system provides on-screen guidance through structured pages. No separate user manual is required for this version.

2.6 Assumptions and Dependencies

- Users have internet access
- Users can understand English
- Supabase services remain available
- The system runs on modern browsers

3. System Features

3.1 Guide System

Provides structured guides for:

- ARC
- Visa
- Housing
- Jobs

3.2 Emergency System

Displays:

- Emergency numbers
- Procedures
- Disaster guidelines

3.3 Help System

Users submit queries stored in database.

3.4 Language System

Provides Korean phrases for daily situations.

4. External Interface Requirements

4.1 User Interface

- Clean UI
- Responsive design
- Korean theme colors

4.2 Software Interface

- Supabase database
- REST API

4.3 Hardware Interface

- Mobile, tablet, desktop

5. Functional Requirements

REQ-1: View Guides

Description: User can access informational pages

Priority: High

Stimulus: User clicks navigation link

Response: System displays relevant guide page

REQ-2: Submit Help Request

Description: User submits query via form

Priority: High

Stimulus: User submits form

Response: Data stored in database and confirmation shown

REQ-3: View Emergency Information

Description: User views emergency details

Priority: High

Stimulus: User opens emergency page

Response: System displays emergency contacts and instructions

REQ-4: Access Language Section

Description: User learns Korean phrases

Priority: Medium

Stimulus: User navigates to language page

Response: System displays categorized phrases

6. Non-Functional Requirements

Performance: Page load time shall not exceed 3 seconds

Usability: Users shall be able to navigate the system without training

Reliability: System uptime shall be at least 95%

Security: User inputs shall be validated before database submission

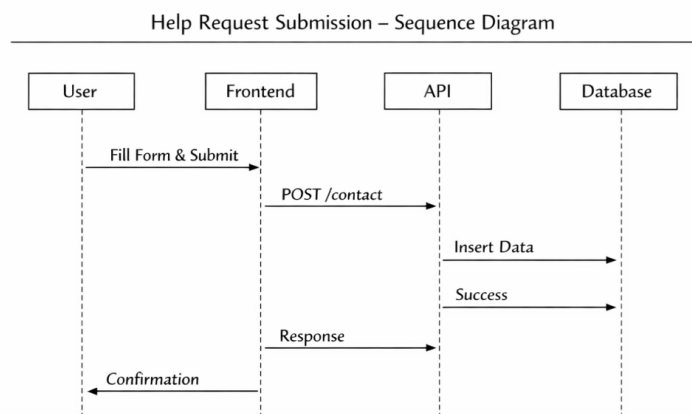
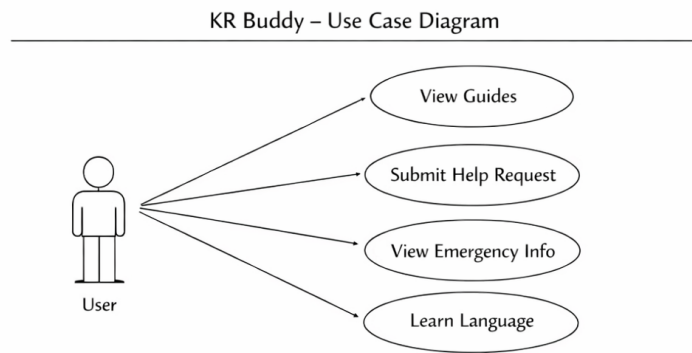
7. Use Case Diagram

Actors:

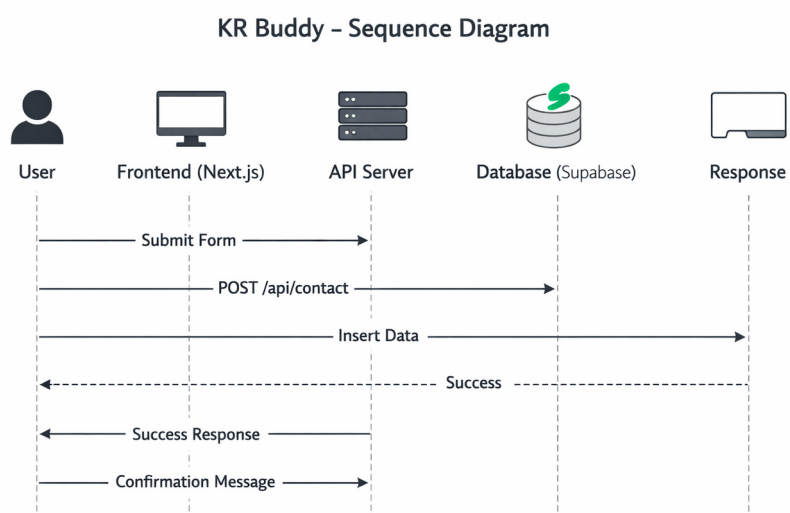
- User

Use cases:

- View Guides
- Submit Help Request
- View Emergency Info
- Learn Language



8. Sequence Diagram



Flow:

User → Frontend → API → Database → Response

9. Database Design

Field	Type
id	bigint
name	text
email	text
category	text
message	text
created_at	timestamp

10. Future Enhancements

- Authentication
- Admin dashboard
- AI chatbot----

11. Conclusion

KR Buddy provides a structured and practical solution for international students in Korea, combining frontend and backend to create a complete full-stack system.

Appendix A: Glossary

ARC – Alien Registration Card

API – Application Programming Interface

User – International student using the system